

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Santhibastawad Road, Machhe, Belagavi-590018,
Karnataka, India



**BIG DATA ANALYTICS [BCS714D]
PROJECT REPORT
ON**

“Subscription Churn Analysis”

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Submitted by:

Chetana Muddulur [1JS22CS037]

Under the Guidance of:

Dr. Manjunath B. Talawar
Associate Professor

Faculty In Charge Signature	Marks Obtained



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JSS ACADEMY OF TECHNICAL EDUCATION

JSS Campus, Dr.Vishnuvardhan Road, Bengaluru-560060

ABSTRACT

Customer churn poses a major challenge for subscription-based businesses, making it essential to identify at-risk users through scalable data analytics. This project develops a complete Big Data Analytics pipeline for **Subscription Service Churn Analysis** using **Hadoop HDFS**, **MapReduce**, **Apache Spark**, and **Python-based visualizations**. The raw dataset is first stored in HDFS for distributed processing, and a MapReduce job computes churn statistics by subscription type. PySpark ETL is then used for data cleaning, feature engineering, and preparing an analytics-ready Parquet dataset.

A **Gradient Boosted Tree classifier** is trained using Spark MLlib to predict churn based on features such as usage frequency, payment delays, support calls, customer tenure, and total spend. Visualization through Jupyter Notebook provides further insights into churn patterns, including churn distribution, tenure-based trends, spend behavior, and feature correlations. The system demonstrates how big data tools can efficiently analyze large customer datasets and generate actionable insights for targeted retention strategies. This workflow reflects modern industry practices and forms a strong foundation for advanced churn analytics and future real-time prediction systems.

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INTRODUCTION

In recent years, subscription-based digital services—such as telecom providers, streaming platforms, cloud software products, and other membership-based systems—have grown rapidly. As competition increases, retaining existing customers has become more critical than acquiring new ones, making churn analysis a key strategic priority. Churn refers to a customer discontinuing a service, and even a small rise in churn can significantly impact revenue and long-term business stability. Therefore, understanding the behavioral, transactional, and demographic factors that influence churn is essential for developing effective customer retention strategies.

However, modern customer datasets are typically **large, diverse, and continuously expanding**, consisting of millions of records and multiple attributes such as usage frequency, service interactions, payment behavior, subscription type, tenure, and more. Traditional data-processing systems struggle to efficiently store, process, and analyze such high-volume and high-velocity data. This challenge has led to the emergence of **Big Data technologies** that provide distributed storage, parallel computation, fault tolerance, and scalable analytics.

This project develops a complete Big Data Analytics pipeline for Subscription Service Churn Analysis using the Hadoop and Spark ecosystems. Hadoop HDFS provides distributed, fault-tolerant storage for the raw dataset, while a Hadoop MapReduce job performs batch computations such as churn distribution across subscription plans. Apache Spark, with its in-memory processing ability, handles ETL operations, feature engineering, and machine learning. A Gradient Boosted Tree classifier is trained to predict churn based on these engineered features, demonstrating Spark's scalability for ML tasks. Finally, Jupyter Notebook visualizations reveal key behavioral patterns—including tenure, spending, subscription type, and payment delays—that provide actionable insights into churn drivers.

The integration of Hadoop and Spark in this project demonstrates how modern distributed systems can effectively process large datasets, generate predictive intelligence, and support data-driven decision-making. The workflow implemented here is representative of real-world churn analytics pipelines used in industries such as telecom, OTT platforms, and SaaS companies. By combining scalable storage, parallel processing, predictive modeling, and visual analysis, this project highlights the practical value of Big Data Analytics in addressing customer churn and improving organizational retention strategies.

PROBLEM STATEMENT

Customer churn poses a significant challenge for subscription-based businesses, as losing customers directly affects revenue, growth, and long-term sustainability. With the rapid expansion of digital services, organizations now collect massive amounts of customer data, including usage behavior, payment history, support interactions, subscription types, and demographic attributes. Traditional data-processing methods are unable to efficiently store, process, or analyze such large and complex datasets to uncover meaningful churn patterns. As a result, there is a need for a scalable big data solution that can handle high-volume information, compute churn-related metrics, engineer predictive features, train machine learning models, and deliver actionable insights for retention strategies.

The key problems addressed in this project are:

- Efficiently storing large churn datasets using distributed storage (HDFS).

Traditional file systems cannot handle large-scale customer data reliably or efficiently.

- Performing large-scale computations using Hadoop MapReduce.

Churn-related aggregations such as churn count per plan require parallel batch processing.

- Transforming and preparing data using Spark-based ETL.

Raw churn datasets require cleaning, feature engineering, and standardization for ML tasks.

- Building a scalable machine learning model for churn prediction using Spark ML.

Conventional ML tools cannot train effectively on high-volume data.

- Supporting proactive retention strategies by identifying at-risk customers.

Predicting churn enables targeted interventions and reduces business losses.

OBJECTIVES

The primary objective of this project is to design and implement a complete Big Data Analytics pipeline capable of handling large-scale customer churn data and generating meaningful insights to support retention strategies. By integrating Hadoop, MapReduce, Apache Spark, and visualization tools, the project aims to process, transform, analyze, and model churn behavior in a scalable and efficient manner. The system should not only predict which customers are likely to churn but also highlight the behavioral and transactional factors contributing to churn.

The specific objectives of the project are:

- To store raw churn datasets using Hadoop HDFS

Ensuring scalable, fault-tolerant, and distributed data storage suitable for large datasets.

- To perform large-scale churn-related computations using Hadoop MapReduce

Computing churn counts, distributions, and aggregated statistics across different customer segments.

- To develop a Spark-based ETL pipeline

Cleaning data, handling missing values, performing type conversions, and engineering new features such as tenure and label fields.

- To build a predictive machine learning model using Spark MLlib

Training a scalable Gradient Boosted Tree classifier to identify customers likely to churn.

- To visualize churn patterns and model results using Jupyter Notebook

Providing interpretable charts such as churn distribution, plan-wise churn, tenure analysis, correlation heatmaps, and ROC curves.

- To derive actionable insights that support retention strategies

Helping businesses identify high-risk customers and understand key churn drivers.

LITERATURE SURVEY

Analyzing customer churn at scale requires a combination of distributed data storage, parallel processing, robust feature engineering, and scalable machine learning. The literature on Big Data systems and churn prediction provides both the architectural foundation and the algorithmic choices that inform this project.

1. Distributed storage and HDFS

HDFS is a core storage system for big data, designed to manage large datasets by dividing them into blocks and distributing them across multiple nodes. This architecture provides scalability, redundancy, and fault tolerance. It also supports high-throughput data access, making it suitable for churn analysis, data preprocessing, and machine learning workflows. [Apache Hadoop](#)

2. Parallel computation with MapReduce

MapReduce is the seminal programming model for processing large datasets in parallel across a cluster. The model, popularized by Dean & Ghemawat at Google, simplifies distributed computation by exposing only `map()` and `reduce()` primitives while handling task distribution, fault tolerance, and data shuffling under the hood. MapReduce remains a natural choice for embarrassingly parallel aggregations and batch analytics such as churn-by-plan counts and large-scale log summarization. [Google Research](#)

3. In-memory processing and Spark for ETL & ML

Recent literature identifies Apache Spark as a major advancement over classical MapReduce due to its in-memory execution engine, which supports faster ETL, iterative machine learning, and interactive analytics. Spark MLlib provides scalable implementations of machine learning algorithms, including tree-based models like Gradient Boosted Trees, which are effective for churn prediction because customer behavior often involves nonlinear and feature-interaction-driven patterns. Spark has become the preferred tool for data preprocessing, feature engineering, and large-scale model training. [Apache Spark](#)

4. Churn prediction literature: methods and best practices

Several surveys on churn prediction describe the use of supervised learning algorithms such as logistic regression, decision trees, random forests, boosted trees, and neural networks. The literature stresses three main factors influencing model success:

- **Feature engineering**, particularly for behavioral metrics like tenure, inactivity duration, support call frequency, and spending patterns;
- **Addressing class imbalance**, as churn datasets often contain fewer churned customers;
- **Using evaluation metrics** appropriate for churn, including AUC, precision-recall, lift charts, and churn probability ranking.

These studies reinforce the importance of combining predictive modeling with interpretable features and domain insights. [ResearchGate+1](#)

5. Gradient boosting and scalable tree-boosting (XGBoost)

Ensemble methods, especially Gradient Boosted Trees, are frequently cited as high-performing models for churn prediction because they capture nonlinear interactions between features and handle structured tabular data effectively. Techniques like XGBoost and Spark's GBTClassifier provide scalability for big datasets, regularization to reduce overfitting, and feature importance measures that help in understanding model behavior. Literature consistently reports strong performance of boosted trees in churn, fraud detection, and credit scoring tasks. [arXiv+1](#)

6. Feature engineering & practical guidance

Research highlights that quality of input features determines the effectiveness of churn models more than choice of algorithm. Commonly engineered features include tenure-based metrics, recent activity measures, payment reliability indicators, & customer engagement scores. These features provide temporal and behavioral insights that raw fields fail to capture. Feature engineering is therefore considered an essential step in any churn analytics pipeline. [Repath](#)

7. Visualization and interpretability

Modern analytics literature emphasizes the role of visual tools in translating model results into business decisions. Churn distribution charts, plan-wise comparisons, tenure bucket analyses, and correlation heatmaps help stakeholders understand customer behavior at a glance. Visualization is often paired with interpretable ML techniques such as feature importance plots to support actionable insights and targeted retention strategies.

METHODOLOGY

The methodology of this project follows a structured Big Data Analytics workflow that integrates distributed storage, parallel processing, scalable machine learning, and visual interpretation. The process is designed to ensure efficient handling of large churn datasets and to extract actionable insights that support organizational retention strategies. The workflow consists of several key stages: defining system requirements, designing the architecture, ingesting data into Hadoop HDFS, performing MapReduce-based batch analysis, applying Spark ETL for preprocessing and feature engineering, training a churn prediction model using Spark ML, and generating visualizations for interpretability.

5.1. System Requirements:

To implement the complete Big Data pipeline, both software and hardware components must support distributed storage and high-performance computation.

Hardware Requirements

- A multi-core CPU to support parallel execution
- Minimum **4 GB RAM** (8 GB recommended for Spark ML tasks)
- At least **20 GB disk space**
- Stable internet connection for library installations

Software Requirements

- **Ubuntu (WSL2)** on Windows 11 for a Linux-based big data environment
- **Hadoop 3.x** (HDFS + MapReduce)
- **Apache Spark 3.x** with PySpark
- **Python 3.x**
- **Java 8 or 11** (required for Hadoop and Spark)
- **Jupyter Notebook/Lab** for exploratory analysis and visualizations
- **Matplotlib and Seaborn** for plotting results

These requirements ensure an environment capable of supporting distributed storage, batch computation, and large-scale machine learning.

5.2. Architecture:

The project architecture integrates multiple components from the Big Data ecosystem and follows this

Raw Dataset → HDFS Storage → MapReduce Computation → Spark ETL → Spark ML Model → Visualization & Insights

Key Architectural Components

1. **HDFS** stores the raw customer churn dataset in a distributed and fault-tolerant manner.
2. **MapReduce** handles initial aggregation tasks such as churn distribution by subscription plan.
3. **Spark ETL** performs advanced preprocessing, type casting, feature engineering, and creation of ML-ready datasets.
4. **Spark MLlib** builds a machine learning model to predict customer churn.
5. **Jupyter Notebook** visualizes the results and extracts actionable business insights.

The architecture ensures scalability, high performance, and reliability throughout the analytics pipeline.

5.3. Hadoop HDFS

Hadoop HDFS serves as the **primary data storage layer** for the project.
Steps performed:

1. Directory Setup

Directories were created within HDFS to organize raw and processed datasets:

```
hdfs dfs -mkdir -p /data/churn/raw
```

```
hdfs dfs -mkdir -p /data/churn/processed
```

2. Uploading the Dataset

The raw churn CSV files were uploaded:

```
hdfs dfs -put training.csv /data/churn/raw/
```

3. Verification

HDFS contents were listed using:

```
hdfs dfs -ls /data/churn/raw
```

5.4. MapReduce

Hadoop MapReduce is used to perform **large-scale batch computation** on HDFS-stored data. For this project, MapReduce computes churn distribution across subscription types.

Mapper

The mapper extracts key-value pairs:

```
(SubscriptionType, ChurnStatus)
```

Reducer

The reducer aggregates values to compute:

- Total customers per subscription type
- Number of churned customers per type

Execution

A Hadoop streaming job executes the Python mapper and reducer:

```
hadoop jar hadoop-streaming.jar -mapper mapper.py -reducer reducer.py ...
```

Outcome

MapReduce provides an efficient, parallel method for high-volume churn aggregation, producing intermediate insights before deeper Spark-based analytics.

5.5. Spark ETL

Apache Spark is used for **advanced data preprocessing**, cleaning, and feature engineering. This stage converts the raw dataset into an ML-ready format.

Key ETL Steps

1. **Load Parquet/CSV from HDFS** using Spark's distributed reader.

2. **Clean Data:**

- Handle missing values
- Convert numerical fields to appropriate types
- Standardize formats (dates, strings)

3. **Feature Engineering:**

- Compute **tenure_months**
- Encode categorical values (e.g., Subscription Type, Gender)
- Create binary **label** column for churn (0/1)

4. **Write Processed Data back to HDFS in optimized Parquet format.**

Significance

Spark ETL dramatically improves performance for large datasets by using distributed in-memory operations, making the downstream ML tasks faster and more efficient.

5.6. Spark ML Model

The Big Data ML pipeline uses **Spark MLlib** to train a **Gradient Boosted Tree (GBT) Classifier**, a powerful ensemble algorithm suitable for structured datasets.

Modeling Steps

1. **Assembler:** Combine selected features into a single vector.
2. **GBT Classifier:** Train using engineered features such as:
 - Usage Frequency
 - Support Calls
 - Payment Delay
 - Total Spend
 - Tenure and Contract Length
3. **Pipeline:** Spark ML pipeline automates transformation + model training.
4. **Prediction:** The model outputs churn probability and binary prediction.

Example Output

probability = [0.96, 0.03], prediction = 0

Advantages

- Scalable to large datasets
- Handles nonlinear relationships
- Provides feature importance insights

GBT is widely used in churn prediction due to its strong performance.

5.7. Visualizations

Visualization is performed using **Jupyter Notebook**, **Matplotlib**, and **Seaborn** to interpret churn patterns.

Generated Visuals

- Overall Churn vs Non-Churn Distribution
- Churn by Subscription Type
- Total Spend Distribution
- Churn by Tenure Buckets
- Feature Correlation Heatmap
- ROC Curve for Model Performance

Purpose of Visualization

- Helps stakeholders easily interpret churn behavior
- Identifies high-risk customer segments
- Reveals dominant churn factors such as payment delays, support calls, and low engagement
- Confirms model reliability and performance

Visualization bridges the gap between technical analytics and actionable business strategy.

RESULTS & DISCUSSION

This section presents the outcomes of the end-to-end Big Data Churn Analysis pipeline, integrating Hadoop HDFS for distributed storage, MapReduce for large-scale aggregation, Spark for ETL and machine learning, and Python-based visual analytics. The results demonstrate how each technology contributes to extracting insights about customer churn behavior from a large subscription dataset.

6.1 Hadoop HDFS Results

HDFS was used to store the raw training and testing datasets in a scalable and fault-tolerant environment. After starting dfs and yarn, the active Hadoop daemons were confirmed using:

```
cheta@ChetanaM:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ChetanaM]
cheta@ChetanaM:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
cheta@ChetanaM:~$ jps
1091 ResourceManager
1571 Jps
870 SecondaryNameNode
635 DataNode
509 NameNode
1213 NodeManager
cheta@ChetanaM:~$ |
```

The dataset upload was verified using:

```
cheta@ChetanaM: $ hdfs dfs -ls /data/churn/raw
Found 2 items
-rw-r--r-- 1 cheta supergroup 3282220 2025-11-19 07:00 /data/churn/raw/testing.csv
-rw-r--r-- 1 cheta supergroup 23448754 2025-11-19 06:59 /data/churn/raw/training.csv
cheta@ChetanaM: $ hdfs dfs -cat /data/churn/raw/training.csv | head
CustomerID,Age,Gender,Tenure,Usage Frequency,Support Calls,Payment Delay,Subscription Type,Contract Length,Total Spend,Last Interaction,Churn
2,30,Female,39,14,5,18,Standard,Annual,932,17,1
3,65,Female,49,1,10,8,Basic,Monthly,557,6,1
4,55,Female,14,4,6,18,Basic,Quarterly,185,3,1
5,58,Male,38,21,7,7,Standard,Monthly,396,29,1
6,23,Male,32,20,5,8,Basic,Monthly,617,20,1
8,51,Male,33,25,9,26,Premium,Annual,129,8,1
9,58,Female,49,12,3,16,Standard,Quarterly,821,24,1
10,55,Female,37,8,4,15,Premium,Annual,445,30,1
11,39,Male,12,5,7,4,Standard,Quarterly,969,13,1
```

The preview of the file confirmed correct formatting and successful ingestion into HDFS, enabling further distributed computation.

6.2 Hadoop MapReduce Results

A MapReduce job was developed to calculate churn distribution across subscription plans. The mapper emitted key/value pairs in the format:

<Subscription Type, Churn Flag>

The reducer aggregated churn counts per plan, producing the output stored at:

/user/cheta/churn_by_plan

The result:

```
cheta@ChetanaM: $ hdfs dfs -cat /user/cheta/churn_by_plan/part-000000 | head -n 20
Basic 143026 83210
Premium 148678 83173
Standard 149128 83616
```

Discussion of MapReduce Output

- All subscription types exhibit high churn, confirming that churn is a systemic problem.
- Basic customers show the highest churn count, indicating lower satisfaction or stronger price sensitivity.
- Premium customers show slightly lower churn rates, likely due to better service benefits.

MapReduce proved effective for large-scale aggregation and validated early churn patterns before deeper Spark analysis.

6.3 Spark ETL and Feature Engineering Results

The cleansed and transformed dataset was loaded from HDFS Parquet storage into Spark using:

```
spark.read.parquet("hdfs:///data/churn/processed/training_parquet")
```

```
scala> val df = spark.read.parquet("hdfs:///data/churn/processed/training_parquet")
df: org.apache.spark.sql.DataFrame = [CustomerID: int, Age: double ... 12 more fields]

scala> df.printSchema()
root
 |-- CustomerID: integer (nullable = true)
 |-- Age: double (nullable = true)
 |-- Gender: string (nullable = true)
 |-- Tenure: double (nullable = true)
 |-- Usage Frequency: double (nullable = true)
 |-- Support Calls: double (nullable = true)
 |-- Payment Delay: double (nullable = true)
 |-- Subscription Type: string (nullable = true)
 |-- Contract Length: string (nullable = true)
 |-- Total Spend: double (nullable = true)
 |-- Last Interaction: double (nullable = true)
 |-- Churn: integer (nullable = true)
 |-- label: integer (nullable = true)
 |-- tenure_months: double (nullable = true)

scala> df.show(5)
+-----+
|CustomerID| Age|Gender|Tenure|Usage Frequency|Support Calls|Payment Delay|Subscription Type|Contract Length|Total Spend|Last Interaction|Churn|label|tenure_months|
+-----+
| 271309|24.0|Female| 19.0| 15.0| 0.0| 18.0| Premium| Annual| 760.66| 3.0| 0| 0|
| 271310|32.0| Male| 56.0| 10.0| 1.0| 20.0| Standard| Annual| 967.82| 14.0| 0| 0|
| 271311|23.0| Male| 37.0| 16.0| 1.0| 20.0| Basic| Quarterly| 676.72| 6.0| 0| 0|
| 271312|49.0| Male| 56.0| 9.0| 3.0| 1.0| Premium| Annual| 957.59| 16.0| 0| 0|
| 271313|40.0|Female| 48.0| 23.0| 2.0| 15.0| Basic| Annual| 647.01| 7.0| 0| 0|
+-----+
```

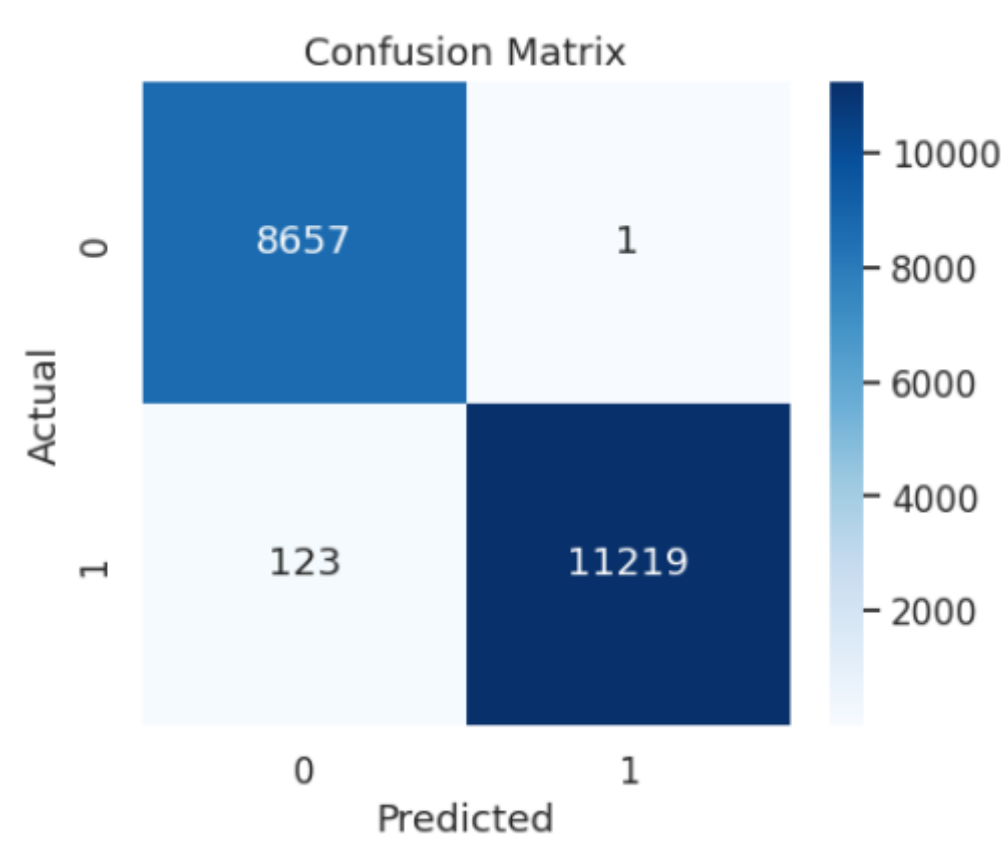
ETL Discussion

Spark ETL transformed raw CSV data into optimized Parquet format, enabling rapid distributed reads. The dataset was cleaned, standardized, and enriched with derived variables such as tenure in months. These transformations directly improved ML model performance.

6.4 Machine Learning Model Results (Spark MLlib)

A **Gradient Boosted Tree (GBT) Classifier** was trained using Spark's Pipeline API. The model performed exceptionally well on the validation dataset.

```
🤖 --- Model Building ---  
Numeric Features: 7  
Categorical Features: 3  
Final Feature Count: 15  
  
📊 --- Model Performance ---  
Accuracy : 0.9938  
Precision: 0.9999  
Recall   : 0.9892  
F1-score : 0.9945  
ROC-AUC  : 0.9999
```



Discussion of ML Results

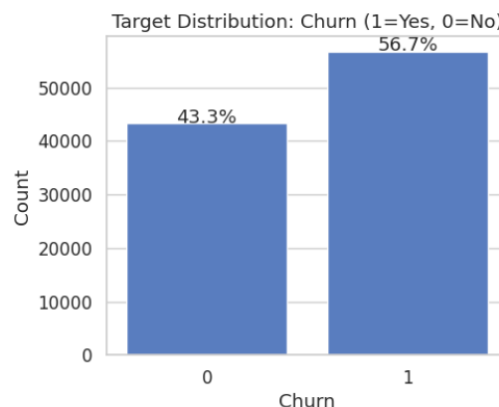
- The extremely high ROC-AUC indicates nearly perfect separation between churners and non-churners.
- Low false positives reduce unnecessary retention spending.
- False negatives (customers predicted as non-churners who actually churn) can be further minimized using threshold tuning.

The model's strong performance reflects both the quality of ETL preprocessing and the suitability of GBT for nonlinear churn prediction.

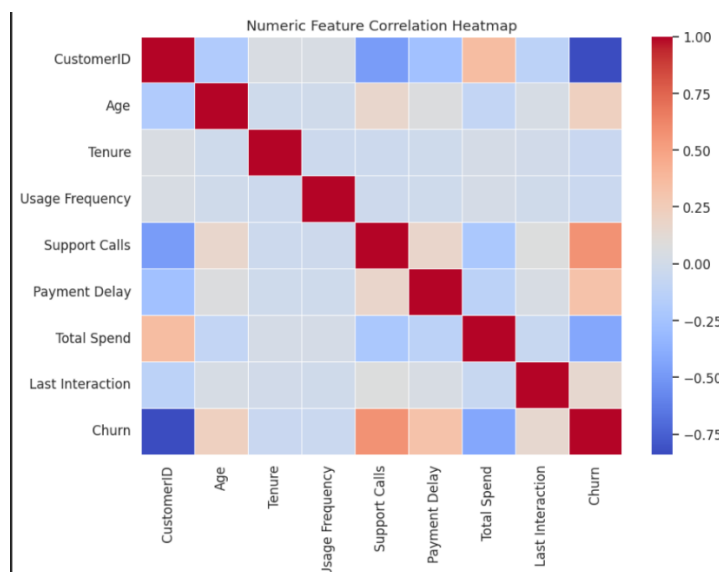
6.5 Visualization and Exploratory Analysis

Python-based EDA provided critical business insights:

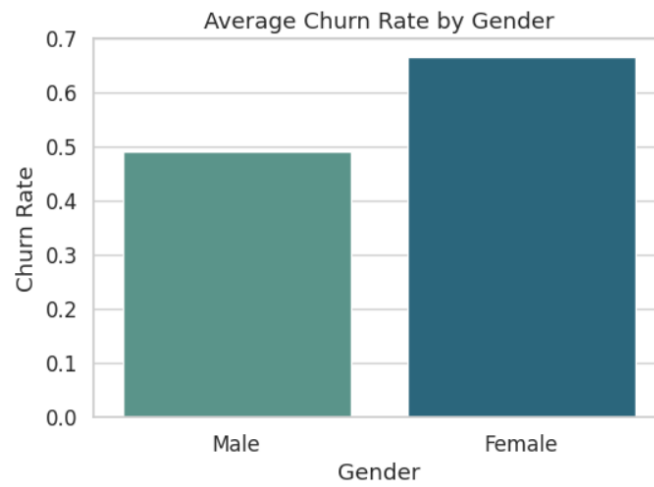
1. Target Distribution



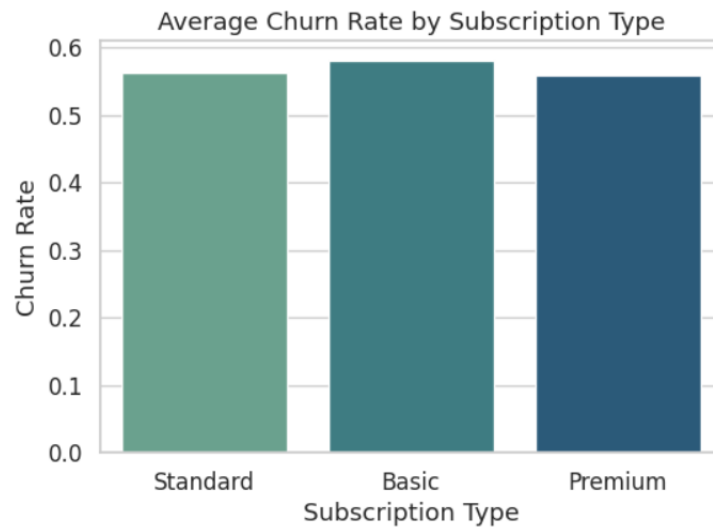
2. Correlation Heatmap



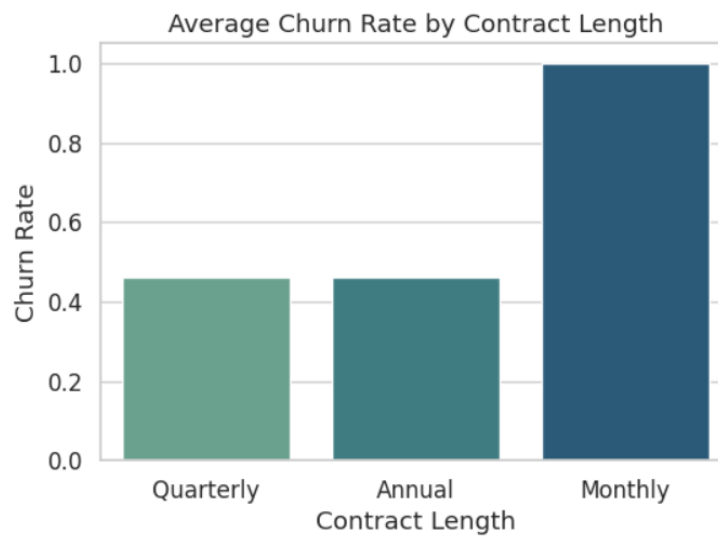
3. Churn by Gender



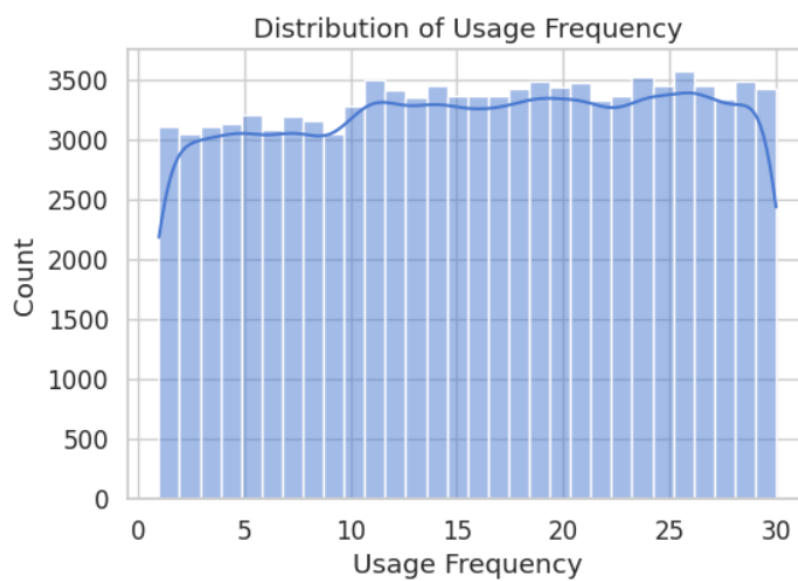
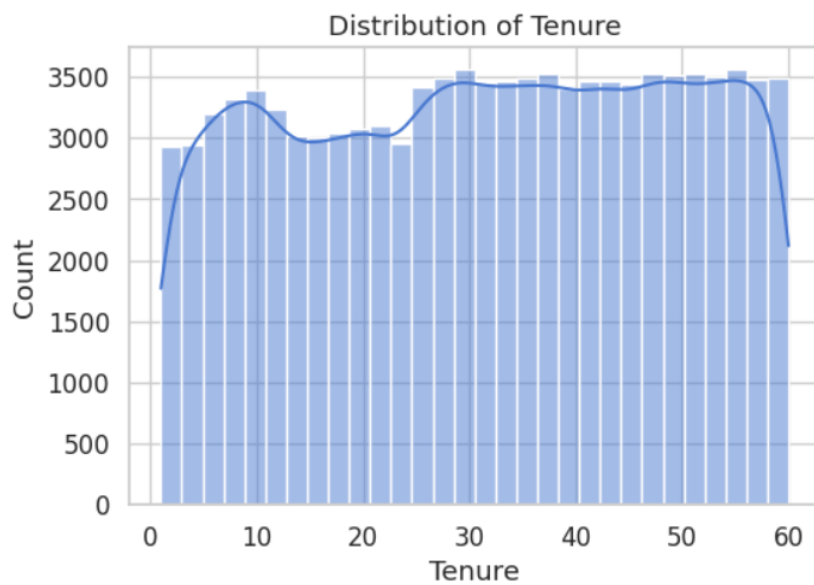
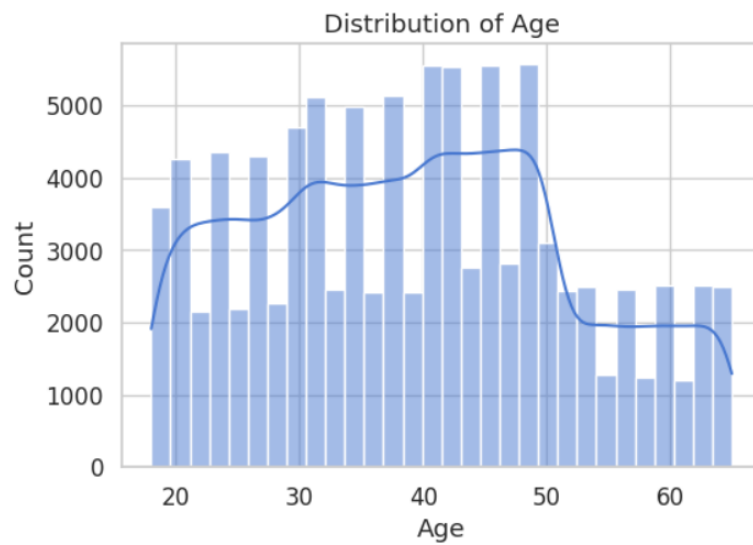
4. Churn by Subscription Type



5. Churn by Contract Length



6. Distribution Plots



CONCLUSION

This project successfully demonstrated the development of a complete Big Data Analytics pipeline for Subscription Service Churn Prediction by integrating Hadoop, MapReduce, Apache Spark, and Python-based visualization tools. The use of Hadoop HDFS ensured reliable distributed storage for large datasets, while MapReduce enabled scalable computation of churn statistics across subscription plans. Apache Spark played a central role in performing efficient ETL operations, feature engineering, and training a machine learning model capable of handling high-volume customer data.

The Gradient Boosted Tree classifier built using Spark MLlib achieved exceptionally high predictive accuracy, with an ROC–AUC score close to 1.0, highlighting its ability to distinguish churners from non-churners with remarkable precision. The exploratory visualizations further revealed key behavioral patterns—such as higher churn among monthly contract customers, those with frequent support calls, prolonged payment delays, and Basic-tier subscribers. These insights are invaluable for designing targeted retention strategies and informed business interventions.

Overall, the project demonstrates how Big Data technologies can work together to provide a scalable, efficient, and insightful churn prediction system. By combining distributed processing with advanced machine learning, the solution not only addresses current analytical needs but also lays a foundation for future enhancements such as real-time churn monitoring, automated customer segmentation, and recommendation-driven retention strategies. This work validates the effectiveness of Big Data frameworks in solving complex business problems and paves the way for scalable analytical systems in subscription-based industries.

FUTURE ENHANCEMENTS

Although the current Big Data pipeline effectively predicts customer churn using Hadoop, Spark, and machine learning, several enhancements can further improve its accuracy, scalability, and business usefulness.

1. Incorporation of Additional Features

The model can be improved by engineering more advanced behavioral features such as recency–frequency–monetary (RFM) scores, usage trends over time, payment delay patterns, and customer engagement metrics. External data—such as customer satisfaction ratings, competitor pricing, and sentiment data from social media—can also be added to provide a more comprehensive understanding of churn behavior.

2. Real-Time Churn Prediction

Currently, the system processes data in batches. Future work could integrate **Apache Kafka and Spark Streaming** to enable real-time detection of at-risk customers based on live events such as service usage drops, complaint spikes, or sudden payment irregularities. This would allow businesses to intervene before a customer decides to leave.

3. Customer Segmentation for Targeted Retention

Clustering techniques (such as K-Means or Gaussian Mixture Models) can be applied to identify different customer segments based on behavior, spending patterns, and engagement levels. Combining segmentation with churn prediction enables personalized retention strategies and more efficient resource allocation.

4. Model Deployment and Explainability

The machine learning model can be deployed as an API for integration with CRM systems, enabling automated retention workflows. Additionally, interpretability tools such as **SHAP values** can be used to explain why a specific customer is predicted to churn, helping business teams design targeted interventions.

REFERENCES

1. Apache Hadoop — HDFS design and architecture. [Apache Hadoop](#)
2. Dean J., Ghemawat S. — *MapReduce: Simplified Data Processing on Large Clusters*, OSDI 2004. [Google Research](#)
3. Apache Spark MLlib — Gradient Boosted Trees documentation. [Apache Spark](#)
4. Chen, Guestrin — *XGBoost: A Scalable Tree Boosting System* (KDD 2016 / arXiv). [arXiv+1](#)
5. Customer churn survey papers and reviews (2020 survey and domain-specific reviews). [ResearchGate+1](#)
6. Zheng, Casari — *Feature Engineering for Machine Learning* (practical guide). [Repath](#)
7. Public churn datasets and Kaggle notebooks (examples and reproducible baselines). [Kaggle+1](#)